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Demonstration of quantum Darwinism on quantum computer

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Abstract
It is well known that environmental decoherence is a crucial barrier in realizing various quantum information processing tasks; on the other hand, it plays a pivotal role in explaining how a quantum system's fragile state leads to a robust classical state. Zurek (Nat Phys 5(3):181–188, 2009) was the first to develop the theory which successfully describes the emergence of classical objectivity of quantum systems using decoherence, introduced by the environment. Here, we consider an n-qubit generalized quantum circuit for the quantum system–environment interaction model, where the first qubit represents the quantum system, and the rest are for the environmental fragments. This quantum circuit is implemented on ibmq_athens and ibmq_16_melbourne for $n = 2, 3, 4, 5, 6$. Its accuracy is checked using quantum state tomography and enhanced using the quantum error mitigation procedure. The reconstructed density matrices are used to investigate quantum-classical correlation and the mutual information between the quantum system and the environment. The investigation proves the quantum Darwinism principle when the quantum circuits are executed on the noise-less simulator; however, it shows the unaccountable behavior when implemented on the real quantum devices. The results via the noise-less simulator successfully prove that the environmental fragment size and the interaction strength play a crucial role in the emergence of classicality.

Keywords IBM quantum experience · QISKit · Quantum Darwinism · Holevo bound · Quantum discord

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1 Introduction

Quantum mechanics is a fascinating but weird theory, inconsistent with the classical one, and its properties explicate its weirdness [1–3]. One of the fundamental properties is superposition, which declares that rather than localize in space (as we see usually), the particle exists simultaneously at different places/states with some probability. For example, one can consider the cat Dead and Alive model proposed by Schrödinger [1]. The superposition property is crucial in defining a quantum system and always exists in the quantum region. The superposition states are highly fragile, and due to this delicate nature, if one interacts with the quantum system, he accidentally destroys its coherence. As a result, it loses its peculiarity and collapses in one of the probable states (pointer state/survival state) [4]. The decoherence theory successfully explains this interaction [5–7]. The famous experiment to observe the decoherence process is the interference effect via electron gun, which has two cases, case 1 and case 2. Case 1: when the bombarding electrons onto slits are entirely isolated, then an interference pattern appears. Case 2: when we decohere the bombarded electrons by observing the path of each electron, consecutively, the interference pattern vanishes, and the observed pattern is the same as via the classical objects.

Decoherence plays a vital role in explaining the rise of classical objectivity of the quantum system [6, 8] and the main drawback in the realization of quantum computers. The quantum systems are very hard to keep isolated from the surrounding environment. So, there is always a leak of information from the quantum system to the environment take place; resultantly, the quantum system loses its peculiarity. However, the decoherence theory simply traced out this leaked information of the quantum system in the environment and treated it inaccessible and irrelevant. Zurek was the first to use this redundant information in the environment to explain the classical objectivity of quantum systems and gave a theory named quantum Darwinism [4, 9]. Quantum Darwinism says that during the environmental decoherence of the quantum system, the information about the quantum system imprints on the environment, and via accessing the environment, one can obtain information about the quantum system's state without interacting with it. And it can explain why we observe the world as classical/robust instead of fragile (as in the quantum realm). It also says that most of the information we extract about the quantum system comes from the environment and term the environment as the communication channel acting between the quantum system and the observer [4, 10, 11].

IBM quantum experience has become a worldwide easily accessible cloud-based quantum computing research platform, where experiments are carried out on real quantum computers of size 1-qubit, 5-qubit, 7-qubit, and 15-qubit. Among these, the researchers mostly use 5-qubit and 15-qubit real devices to perform their quantum computational and quantum informational tasks. IBM also provides a python module named quantum information software kit for quantum computation (QISKit) [12], where one can write programs, for creating and executing the quantum circuits on real devices and collect the execution results. A large number of tasks have already been performed such as quantum simulation [13–16], quantum machine learning [17, 18], quantum algorithms [19–21], quantum communication [22–24], quantum cryptography [25], quantum circuit architecture [26–28], quantum game [29, 30], to name a few.

Recently, on a photonic quantum simulator [16, 31] the emergence of classical objectivity from the quantum system is tested using the quantum Darwinism state by Chen *et al.* [32]. They had considered a six photonic system in which one photon is used as the quantum system, and the rest is for the environment. They have discussed the obtained quantum-classical correlation and mutual information between the quantum system and environment for two different angle parameters representing the interaction strength.

In addition to the Chen *et al.*'s work, the contributions of this article are listed as follows:

1. We design a generalized quantum circuit to calculate the quantum-classical correlation and mutual information present between the environment and the interacted quantum system based on the quantum Darwinism technique on IBM's superconducting quantum devices named **ibmq_athens** and **ibmq_16_melbourne**.
2. The generalized quantum circuits consist of two different quantum registers, one representing a quantum system and the other for the environment. In this paper, the quantum register for the quantum system have one qubit, and the quantum register for the environment have qubits that range from one to five. So, in total, we have five quantum circuits.
3. We have performed the quantum state tomography to analyze the quantum state fidelity, purity, mutual information, Holevo quantity, and quantum discord.
4. Finally, the quantum error mitigation is performed on the executed results to enhance the desired state accuracy. The corresponding mitigated outcomes are presented in front of the without mitigated ones.

The rest of the paper is organized as follows. In Sect. 2, we briefly discuss the quantum Darwinism. The circuit formation for quantum Darwinism state with the execution on real quantum devices, quantum state tomography, and quantum error mitigation is presented in Sect. 3. In Sect. 4, we present the calculation of the mutual information, Holevo quantity, and quantum discord and then a discussion on their results and plots. Finally, we conclude in Sect. 5.

2 Quantum Darwinism

Based on scientific experiments, it is well known that the fundamental and extraordinarily tiny particles are building blocks of the universe. These particles obey quantum properties (superposition, entanglement, etc.). However, we cannot observe these quantum properties in the physical world; we see the whole world as classical (violating quantum properties). So where does this all quantumness go? Why cannot we observe it? And instead of seeing fragility, why do we see robustness? The detailed explanation of these types of riddles lies in the quantum Darwinism principle.

Figure 1 shows a quantum system, environmental fragments, and different independent observers (independent means there is no pre-agreement between the observers regarding measurement). There is a well-suitable condition for the environmental par-

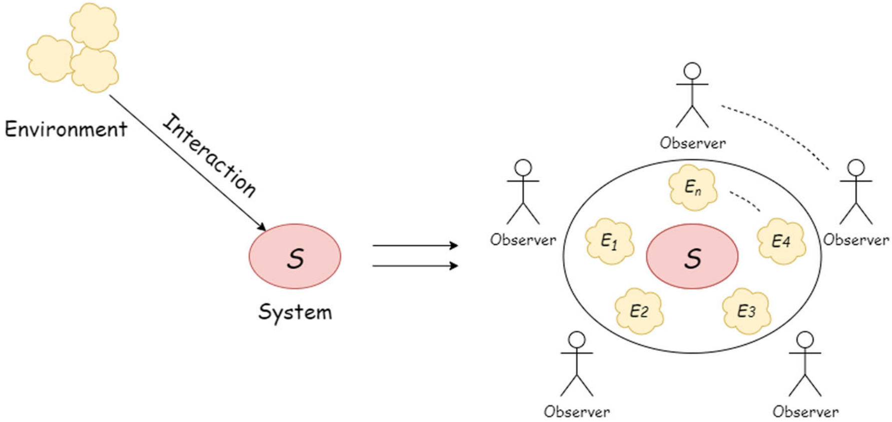

Fig. 1 The figure shows the interaction between the quantum system and the environment. This interaction forms an ensemble where the environment surrounds the quantum system. This ensemble is then observed by the independent and with no prior consensus observers who can only access the environment. After the agreement made by the observers on the outcomes of these observations, classical objectivity arises

ticles, in which no interaction allows with each other. Because if they interact with each other, the interaction will mix the quantum system's imprinted information, making it harder to measure correctly. So, when the non-interacting particles of the environment interact with the quantum system, they work like separate blank sheets, where the quantum system can imprint information.

After the quantum system–environment interaction, the environmental particles witness the quantum system's state, and observing these particles will reprepare the quantum system's state. The quantum system should imprint multiple redundant copies of information on the environmental particles, which is essential for the objectivity [33]. Because observers cannot observe the whole environment, only tiny environmental fragments are accessible. The observers simultaneously measure the different environmental fragments from various positions and agree on the outcome information. As a result, the quantum system's objectivity arises, which indicates classicality. The states that come forward in this observation procedure are known as pointer states. Pointer states are the selected stable states by the environment that have several redundant copies compared to other possible states on the environment and survive during the interaction with the environment (means did not get mixed in the environment) till they are observed.

Now, the question that arises is how much information does an environmental fragment contain about the quantum system? Or what size of the fragment have all of the information about the quantum system? The mutual information between the quantum system and the environmental fragments answers these questions.

$$I(S : E_i) = H_S + H_{E_i} - H_{SE_i} \tag{1}$$

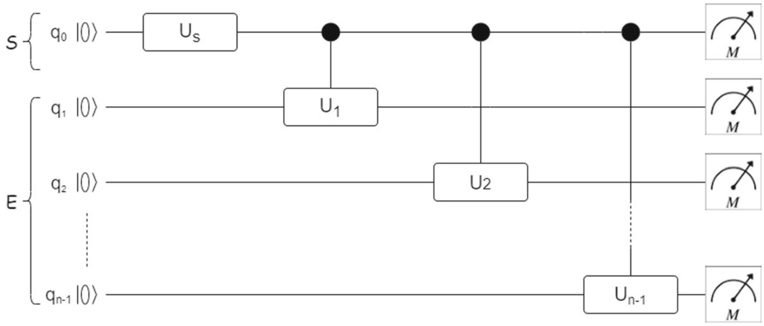

Fig. 2 The figure shows the quantum circuit of n-qubit, one single-, and (n-1) two-qubit quantum gates provided by IBM QX to produce a quantum Darwinism state. We split these n-qubit into two different quantum resisters as the first resister represents the quantum system, and the rest (n-1) is for the environment

Here H is the von Neumann entropy of the reduced density matrices of the quantum system S, environmental fragment E_i, and combined quantum system and environmental fragment SE_i.

3 Experimental realization of Darwinism states in IBM Q devices

This section discusses the execution of the quantum Darwinism principle on two different real quantum devices, the 5-qubit “ibmq_athens" and the 15-qubit “ibmq_16_melbourne" available on the IBM cloud. Figure 2 shows the quantum circuit that we have used in the experiments. The initial state of the circuit is $|00\ldots0_n\rangle$, and after applying the unitary gate (U_S) on the first qubit and then a series of control unitary (cU_i) operations as shown in Fig. 2, it encodes the initial state into the quantum Darwinism state given by,

$$|\psi_N\rangle = \alpha|0\rangle_S \otimes_{i=1}^{N} |0\rangle_i + \beta|1\rangle_S \otimes_{i=1}^{N} \left(\cos\frac{\theta_i}{2}|0\rangle_i + \sin\frac{\theta_i}{2}|1\rangle_i\right) \tag{2}$$

Here α and β are the normalization constants, and N is the number of environmental qubits. Furthermore, the Qiskit tool is used to perform multiple tasks that are discussed and presented in the rest paper.

3.1 Circuit formation

The quantum circuit for quantum Darwinism state includes $U_S = U(\theta, 0, 0)$, $cU_i = cU(\theta_i, 0, 0, 0)$ quantum gates and quantum and classical registers, which are available on the IBM QX platform [34].

$$U(\theta, \phi, \lambda) = \begin{pmatrix} \cos\frac{\theta}{2} & -e^{\iota\lambda}\sin\frac{\theta}{2} \\ e^{\iota\phi}\sin\frac{\theta}{2} & e^{\iota(\phi+\lambda)}\cos\frac{\theta}{2} \end{pmatrix} \tag{3}$$

Table 1 Interaction strength for each quantum circuit

Case	Interaction strength A	Interaction strength B
2-qubit	π	$\frac{2\pi}{5}$
3-qubit	π, π	$\frac{2\pi}{5}, \frac{5\pi}{9}$
4-qubit	π, π, π	$\pi, \frac{2\pi}{5}, \frac{5\pi}{9}$
5-qubit	π, π, π, π	$\pi, \pi, \frac{2\pi}{5}, \frac{5\pi}{9}$
6-qubit	π, π, π, π, π	$\pi, \pi, \pi, \frac{2\pi}{5}, \frac{5\pi}{9}$

$$cU(\theta, \phi, \lambda, \gamma) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & e^{\iota\gamma}\cos\frac{\theta}{2} & -e^{\iota(\gamma+\lambda)}\sin\frac{\theta}{2} \\ 0 & 0 & e^{\iota(\gamma+\phi)}\sin\frac{\theta}{2} & e^{\iota(\gamma+\phi+\lambda)}\cos\frac{\theta}{2} \end{pmatrix} \tag{4}$$

- First, we create the quantum system's state for our experiment via applying the U_S gate on the first qubit (q_0). The U_S gate generates quantum state of form $\alpha|0\rangle + \beta|1\rangle$, where $real(\alpha) = real(\beta)\tan(\frac{\theta}{2})$. Hence, one can manipulate values of α and β by changing θ, and generate a desired quantum system's state. In this paper we have taken $\theta = \frac{\pi}{2}$ and the phase term is excluded from our experiment.

$$|00\ldots 0_n\rangle \xrightarrow[\text{first qubit}(q_0)]{\text{apply}U_S\text{on}} \alpha|00\ldots 0_n\rangle + \beta|10\ldots 0_n\rangle \tag{5}$$

- Now, we apply cU_i gates with the quantum system's qubit (q_0) as control and the environmental qubits as target as shown in Fig. 2. After these operations, the quantum system's qubit is entangled with the environmental qubits. The interaction strength of this entanglement is given by the angle θ_i in the cU_i operation.

$$|00\ldots 0_n\rangle + |10\ldots 0_n\rangle \xrightarrow[\text{shown in Fig. 2}]{\text{apply}cU_i\text{as}} |\psi_N\rangle \tag{6}$$

Table 1 shows the values of θ_i for each experiment in this paper. Also, note that we have taken two different interaction strengths A and B for each experiment.

3.2 Quantum state tomography

Quantum state tomography is a method to reconstruct the state of a quantum circuit based on the probabilistic outcomes by measuring it in all possible combinations of Pauli basis (I, X, Y, Z) [35]. It is to be noted that all the measurements are done in Z-basis on the IBM Q platform. Hence, to measure in X and Y basis, we need to apply H and $S^\dagger H$, respectively, before the Z-basis measurement box. And for the I-basis measurement, we remove the measurement box from the qubit line.

From this tomography technique, we collect the state in the form of a density matrix, also called the reconstructed experimental density matrix. The expression for the experimental reconstructed density matrix is given as (for n-qubit systems),

$$\rho^E = \frac{1}{2^n} \sum_{t_1,t_2,\ldots,t_n=0}^{3} (S_{t_1} \otimes S_{t_2} \otimes \cdots \otimes S_{t_n}) \cdot \sigma_{t_1,t_2,\ldots,t_n} \tag{7}$$

where $\sigma_{t_1,t_2,...t_n} = (\sigma_{t_1} \otimes \sigma_{t_2} \otimes \cdots \otimes \sigma_{t_n})$, and $\sigma_{t_1}, \sigma_{t_2}, \ldots, \sigma_{t_n}$ represent Pauli matrices. Here, the indices $t_1, t_2, \ldots, t_n$ can take values 0, 1, 2, 3 corresponding to I, X, Y, Z. And, S is the Stokes parameter, which is defined as $S = P_0 - P_1$ for X, Y, Z basis, and $S = P_0 + P_1$ for I basis. Here, we use X, Y, and Z basis for measurement and did not include the I basis measurement because we can calculate the Stoke parameters value for measurement setting IO and OI from OO measurement settings outcome as [35],

$$\begin{aligned} S_{OO} &= (P0 - P1) \otimes (P0 - P1) = P00 - P01 - P10 + P11 \\ S_{IO} &= (P0 + P1) \otimes (P0 - P1) = P00 - P01 + P10 - P11 \\ S_{OI} &= (P0 - P1) \otimes (P0 + P1) = P00 + P01 - P10 - P11 \end{aligned} \tag{8}$$

Here, O can be X, Y, and Z basis measurements. Hence, using OO measurement basis probabilities, we can calculate the Stokes parameters, S_{IO} and S_{OI}, by changing the sign. Thus, in total, we have 3^n different settings for the n-qubit quantum circuit, for 2-, 3-, 4-, 5-, and 6-qubit quantum circuits; we have 9, 27, 81, 243, and 729 measurement settings used to perform quantum state tomography, respectively.

We reconstruct the experimental density matrices using execution results from the real quantum devices, the 15-qubit ibmq_16_melbourne, and the 5-qubit ibmq_athens with maximum shots for each setting (shots=8192). A single shot represents a single measurement performed for a particular quantum circuit. The quantum state tomography is also used for calculating the closeness between theoretical and experimental density matrices. This closeness is called fidelity (F), and its expression is given as,

$$F(\rho^T, \rho^E) = Tr\left(\sqrt{\sqrt{\rho^T}\rho^E\sqrt{\rho^T}}\right) \tag{9}$$

where $\rho^T = |\psi\rangle\langle\psi|$ is the theoretical density matrix of the quantum circuit and ρ^E is the experimental density matrix shown in Eq. (7).

In Tables 2 and 3 we present the fidelity and purity of ρ^E (with and without mitigation) for 2-, 3-, 4-, 5-, and 6-qubit systems after executing the quantum circuits. Note that in each of the cases, after performing the mitigation, the fidelity and purity both are increased. (It means that after performing the mitigation process, the obtained density matrix is more close and pure than without mitigation.) Also, the fidelity and purity of the density matrix obtained from ibmq_athens are much higher than ibmq_16_melbourne for the five-qubit system, and for the rest, it is a bit higher. However, overall, there is a decrease in the state's fidelity and purity as we go from 2- to 6-qubit system for both devices. The main reason is that as we increase the number of qubits in the quantum circuits, the number of measurement settings exponentially increases in the order of 3^n. We know that each execution result contains some error,

Table 2 Fidelity results obtained from the real devices ibmq_16_melbourne and ibmq_athens, with and without mitigation with interaction strength parameters A and B

Interaction strength	Case	Athens	Mitigated Athens	Melbourne	Mitigated Melbourne
	S-E_1	0.9427	0.9916	0.8773	0.9855
	S-E_2	0.9143	0.9821	–	–
A	S-E_3	0.7419	0.7933	0.6069	0.9681
	S-E_4	0.7260	0.8101	0.2210	0.3533
	S-E_5	–	–	0.0557	0.0990
	S-E_1	0.9568	0.9751	0.9276	0.9663
	S-E_2	0.9261	0.9596	–	–
B	S-E_3	0.7603	0.7412	0.5667	0.7658
	S-E_4	0.7260	0.7915	0.1318	0.1848
	S-E_5	–	–	0.0727	0.0896

Here, S-E_n represents 1-quantum system qubit and n-environmental qubits interaction

Table 3 Purity of the density matrix obtained from the real devices ibmq_16_melbourne and ibmq_athens, with and without mitigation with interaction strength parameters A and B

Interaction strength	Case	Athens	Mitigated Athens	Melbourne	Mitigated Melbourne
	S-E_1	0.8924	0.9854	0.7861	0.9852
	S-E_2	0.8479	0.9747	–	–
A	S-E_3	0.6140	0.7167	0.4319	1
	S-E_4	0.6086	0.7518	0.1988	0.5622
	S-E_5	–	–	0.1346	0.4035
	S-E_1	0.9236	0.9630	0.8696	0.9458
	S-E_2	0.8736	0.9394	–	–
B	S-E_3	0.6891	1	0.4736	1.2
	S-E_4	0.6126	0.7226	0.2253	0.5242
	S-E_5	–	–	0.1294	0.3370

Here, S-E_n represents 1-quantum system qubit and n-environmental qubits interaction

consequently accommodating an exponential increase in error while performing quantum state tomography. We have transpile all our quantum circuits with a higher level of optimization in each measurement setting before executing on the real devices to obtain the experimental states with maximum fidelity.

4 Mutual, quantum, and classical information

4.1 Mutual information

The mutual information is the difference of uncertainty between quantum system (S) and environmental fragment (E_i) separately and combined (SE_i). Equation (1) quantifies the mutual information occupied by the quantum system and environmental fragment.

1. When $I(S : E_i) = 1$, E_i has the complete record of the quantum system (S).
2. When $I(S : E_i) < 1$, E_i does not have the complete record of the quantum system (S).
3. When $I(S : E_i) > 1$, then
   - E_i is the whole environment.
   - E_i has more information than the quantum system alone.

In our experiment, we have successfully reconstructed the density matrix from the execution results of each quantum circuit. Then the mutual information using Eq. (1) and density matrices ρ^T, ρ^E, and mitigated ρ^E, is calculated and plotted in Fig. 3. The quantum error mitigation procedure helps in getting closer to the desired result from the executed ones. From the figure, we notice that the MI results for ρ^T are identical according to the theory [4]. However, the executed or real device gives satisfactory results for the lower number of qubit systems (2- and 3-qubit systems). And, as we increase the number of environmental qubits (for 4-, 5-, and 6-qubit systems), The value of MI becomes unsatisfactory and unexplainable. From the MI theory, it is clear that the value of MI is always positive [36], but here we obtain some negative values too. These odd results are due to the noise and error introduced during calculating the density matrices, as discussed in the quantum state tomography part. Also, note that while interaction strength is $\frac{2\pi}{5}$ or $\frac{5\pi}{9}$, the obtained information is lesser than when it is π. Because, with π, the environmental qubits highly correlate with the quantum system qubit (q_0).

4.2 Holevo quantity

The Holevo quantity χ is the measurement of the classical information passed by the quantum channel (here environment E) about the observable (here quantum system S), and this quantity is calculated as

$$\chi(S, E_i) = H\left(\sum_s p_s \rho_{E_i|s}\right) - \sum_s p_s H(\rho_{E_i|s}) \tag{10}$$

Here, $\rho_{E_i|s}$ represents the density matrix of the environmental fragment conditioned on outcome of the quantum system's states with probability p_s. Here, we calculate this value as,

$$|\psi_N\rangle = \alpha|0\rangle_S \otimes_{i=1}^N |0\rangle_i + \beta|1\rangle_S \otimes_{i=1}^N \left(\cos\frac{\theta_i}{2}|0\rangle_i + \sin\frac{\theta_i}{2}|1\rangle_i\right) \tag{11}$$

Here, the first qubit shows the quantum system and the rest are for the environment, and the state becomes as follows,

$$\alpha|0\rangle_s|\psi_{E_i|0}\rangle + \beta|1\rangle_s|\psi_{E_i|1}\rangle \tag{12}$$

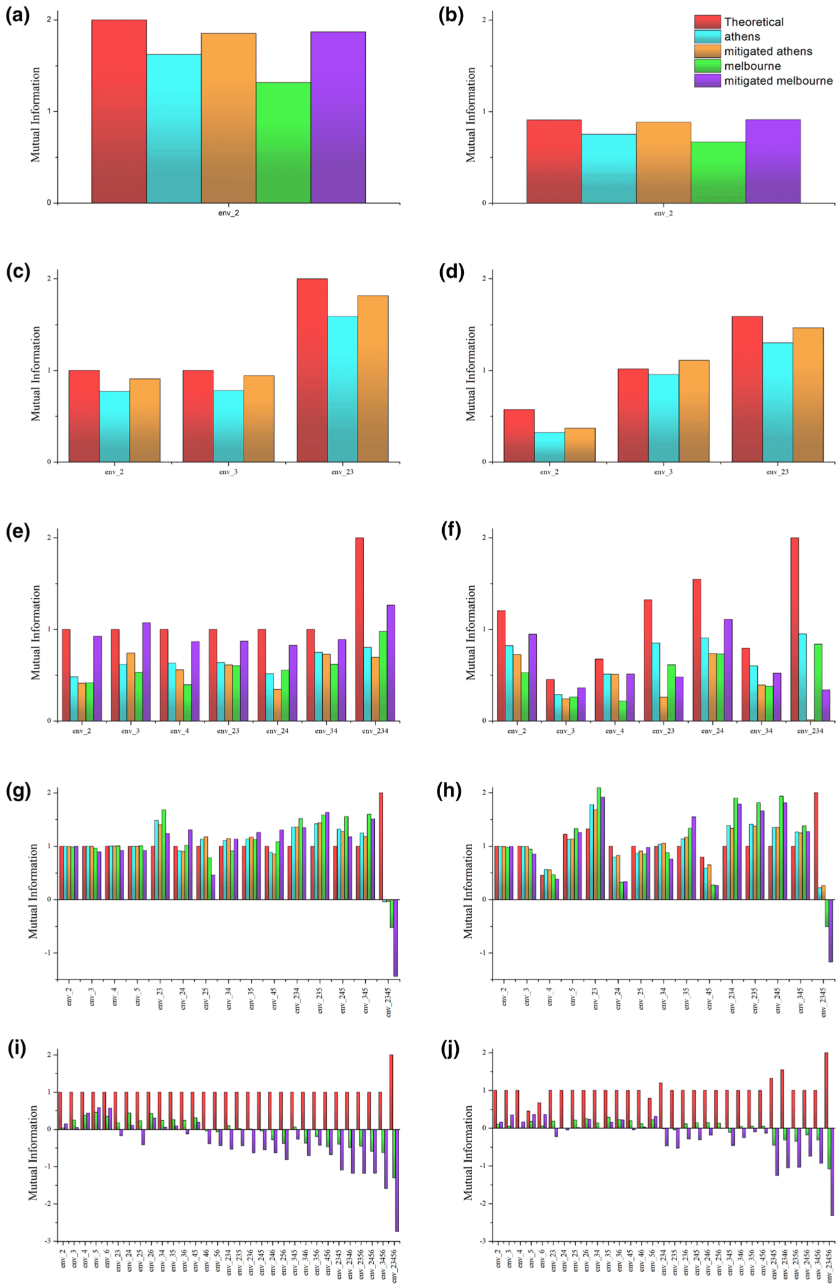

Fig. 3 Mutual information (MI) plots between the quantum system and different environmental fragments for the 1-, 2-, 3-, 4-, and 5-qubit as the environment are given in **a**, **b**, **c**, **d**, **e**, **f**, **g**, **h**, and **i**, **j**, respectively. We have plotted the interaction strength A and B results on the left and right columns, respectively. In red, theoretical; in cyan and orange, Athens with and without mitigation, respectively; in green and violet, Melbourne with and without mitigation, respectively; the MI results are shown (Color figure online)

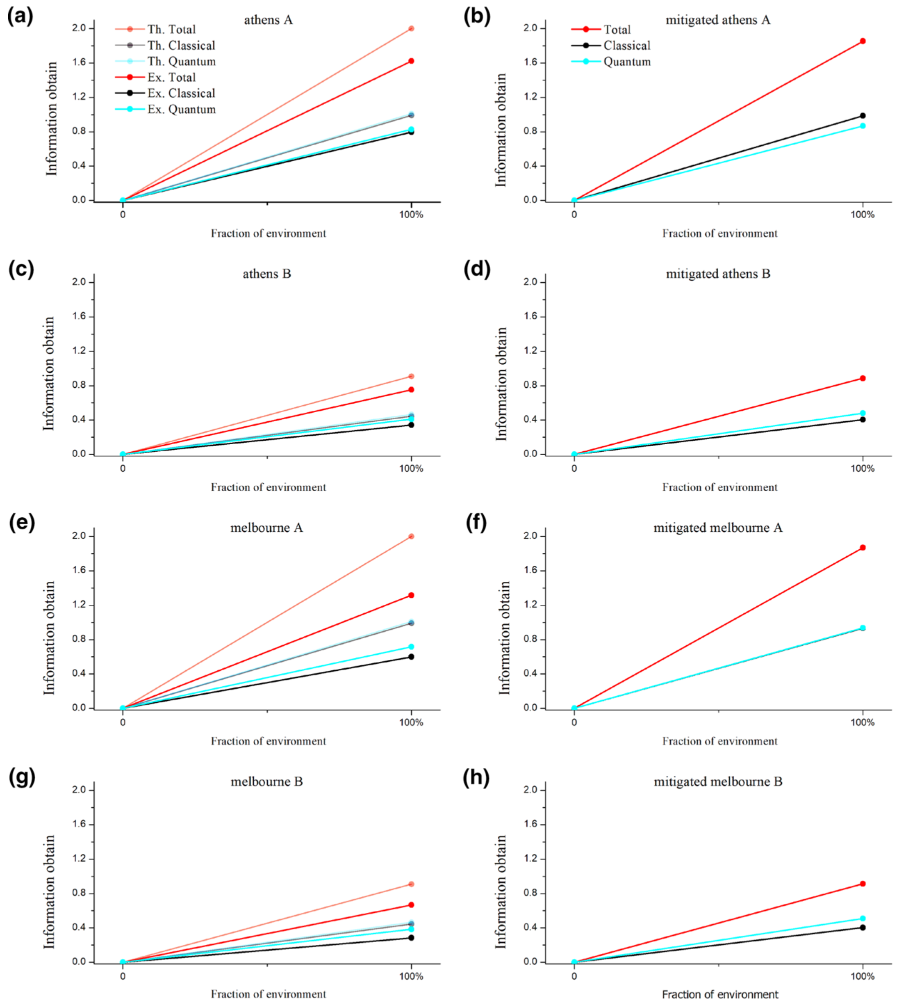

Fig. 4 The figure compares obtained information for 1- qubit quantum system and 1-qubit environment. The title of each plot indicates the name of the real device and interaction strength parameter. The plots in the right column are for mitigated results and in the left column are for the theoretical and experimental results

and the density matrix is $\rho_{E_i|0} = |\psi_{E_i|0}\rangle\langle\psi_{E_i|0}|$ and $\rho_{E_i|1} = |\psi_{E_i|1}\rangle\langle\psi_{E_i|1}|$. Figures 4, 5, 6, 7, and 8 show the obtain quantity $\chi(S, E_i)$ from each quantum circuit for density matrices ρ^T, ρ^E, and mitigated ρ^E.

4.3 Discord

Quantum discord is the measurement of quantumness of correlation, and it generally appears when the involved systems are quantum. It quantifies by calculating the

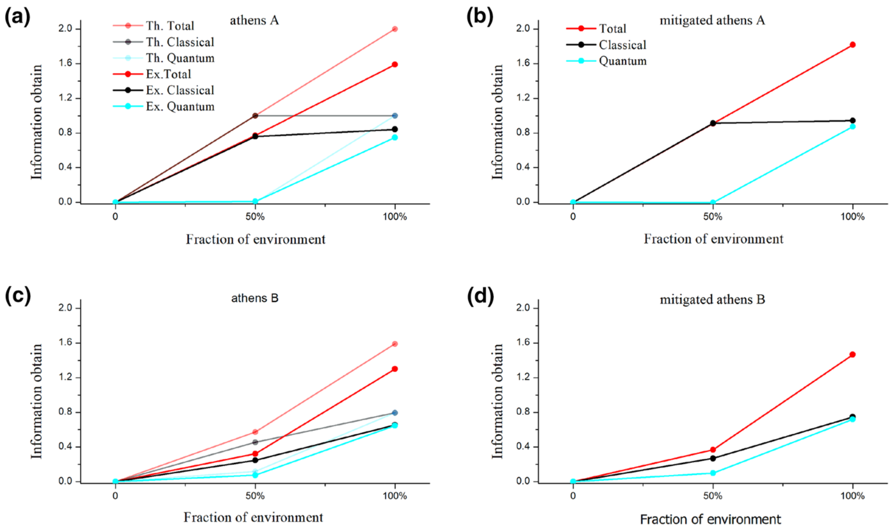

Fig. 5 The figure compares obtained information for 1- qubit quantum system and 2-qubit environment. The title of each plot indicates the name of the real device and interaction strength parameter. The plots in the right column are for mitigated results and in the left column are for the theoretical and experimental results

difference between Eqs. (1) and (10).

$$D(S, E_i) = I(S, E_i) - \chi(S, E_i) \tag{13}$$

Note that the quantities calculated in Eqs. (1) and (10) are different in quantum case. It can be easily seen by definition and expression given in Eq. (13) of quantum discord.

The quantum discord (D) and Holevo quantity (χ) plots are correlated [37], such as if the Holevo plot rises to the classical plateau early as we select the minimal fragment of the total environment, then discord only rises when we select the whole environment. As we observe in the interaction strength A case (Figs. 5a, 6a, g, 7a, g, and 8a see light black and light cyan line), and if we see delayed rise in the Holevo quantity toward the plateau, then consecutively discord starts rising earlier as in the interaction strength B case (Figs. 5c, 6c, i, 7c, i, and 8c) see light black and light cyan line.

4.4 Discussion on obtained results and plots

Figures 4, 5, 6, 7, and 8 show the difference in information obtained from theoretical, real device and mitigation of real device's outcomes. The left column in these figures shows theoretical and experimental results, whereas the right column shows the mitigated results. Here, light red, light black, and light cyan lines represent theoretical total/mutual, classical, and quantum correlations. And, the dark red, dark black, and dark cyan lines represent experimental and mitigated total/mutual, classical, and

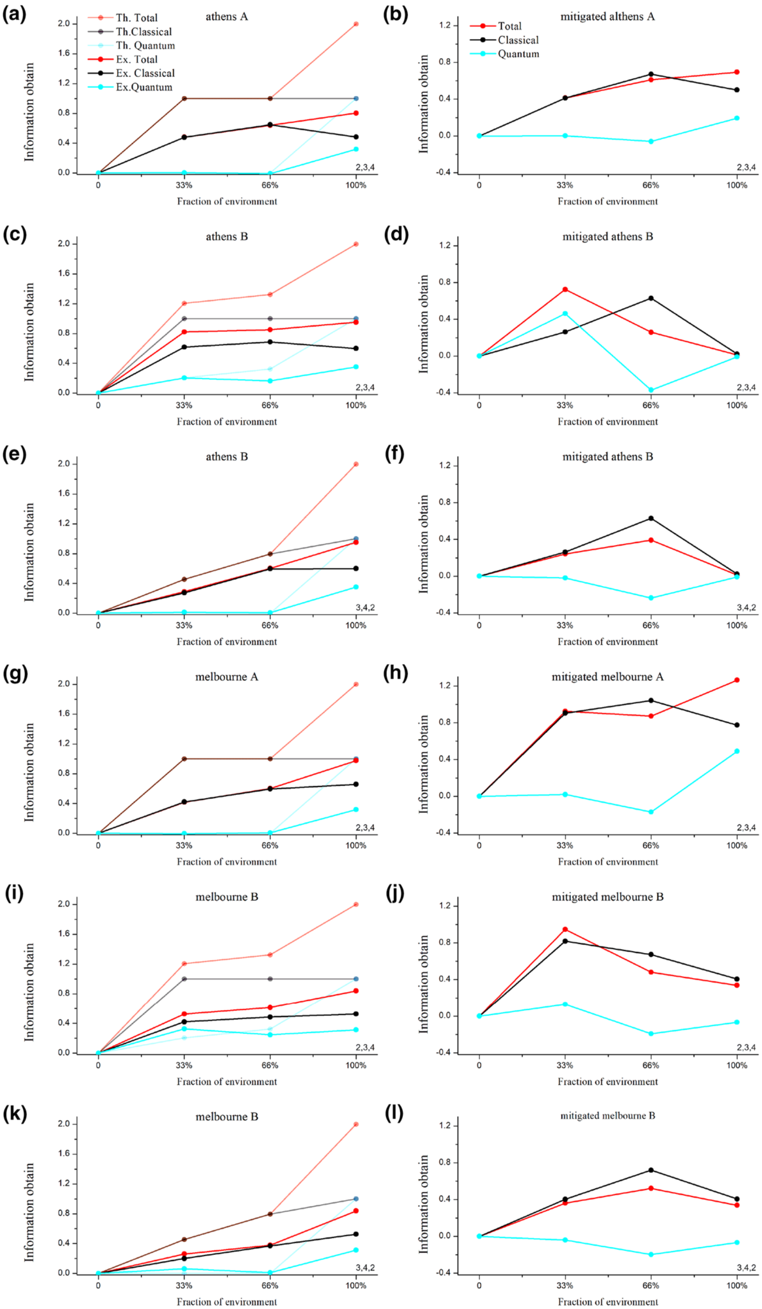

Fig. 6 The figure compares obtained information for 1- qubit quantum system and 3-qubit environment. The title of each plot indicates the name of the real device and interaction strength parameter. The plots in the right column are for mitigated results and in the left column are for the theoretical and experimental results

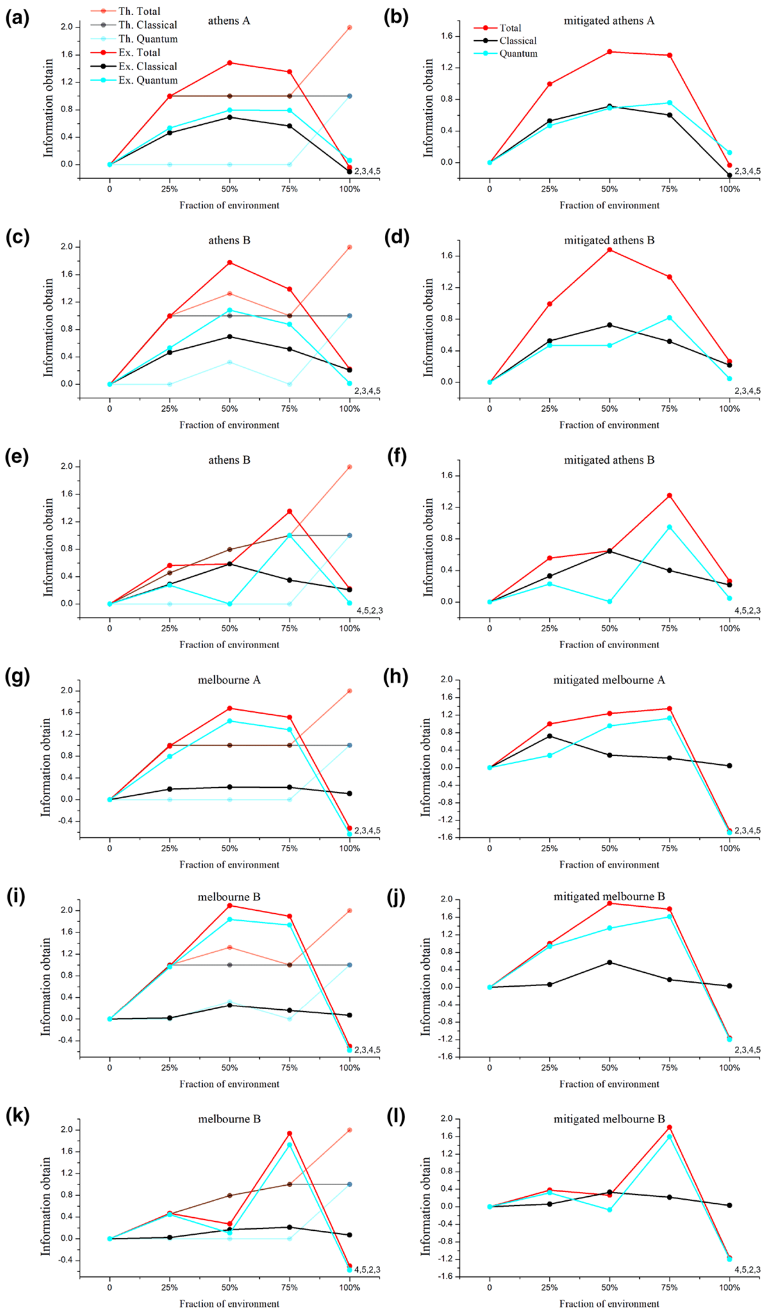

Fig. 7 The figure compares obtained information for 1- qubit quantum system and 4-qubit environment. The title of each plot indicates the name of the real device and interaction strength parameter. The plots in the right column are for mitigated results and in the left column are for the theoretical and experimental results

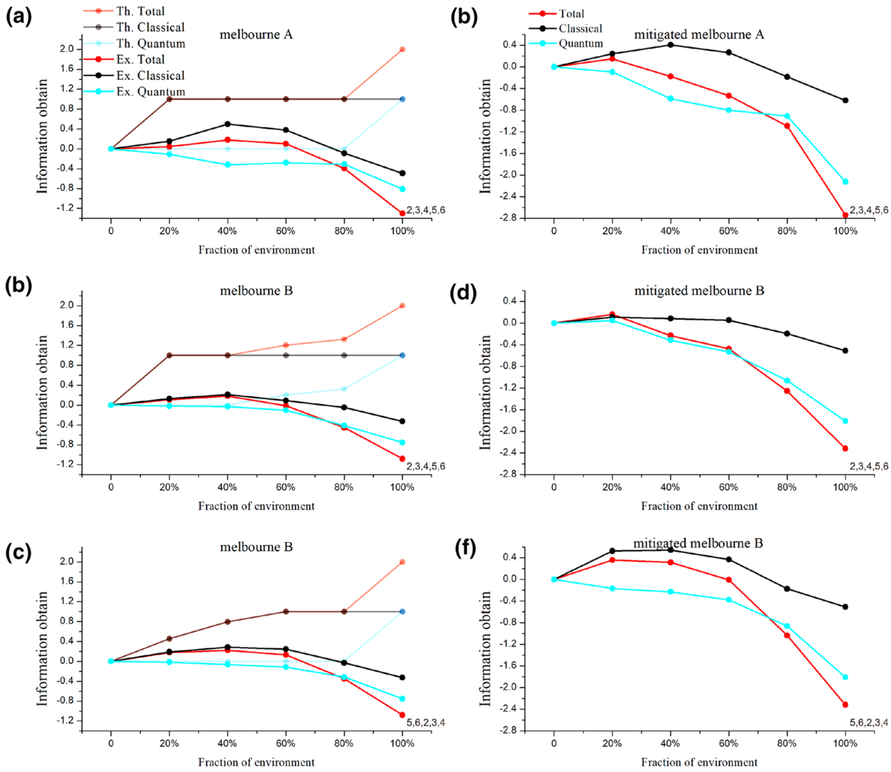

Fig. 8 The figure compares obtained information for 1- qubit quantum system and 5-qubit environment. The title of each plot indicates the name of the real device and interaction strength parameter. The plots in the right column are for mitigated results and in the left column are for the theoretical and experimental results

quantum correlations between the quantum system and the environment. The X- and Y-axes show the fragment of environment and obtained information, respectively, and the title label of plots is divided such that {device name} {interaction strength} (Ex:-Athens B). The theoretical results are identical with the quantum-classical correlation theory [38–40], which says that the classical information can be achieved by accessing some fragment of the environment; however, the quantum information can be observed only when the fragment is the whole environment. Moreover, the experimental results are not equivalent to the theoretical because the execution on the real device accommodates noise in the execution, resulting in a reduction in fidelity and purity of the reconstructed quantum state. This reduction leads to the variation in the obtained information imprinted on the environment. This variation can be seen and compared from Fig. 3 (in mutual information), and Figs. 4, 5, 6, 7, and 8 (in classical-quantum correlation). Here, the order of environmental fragments in Figs. 6, 7 and 8 is from 0 to 100% following the order as written in the plot on right hand side on the X-axis. For example, if we written the order is (a, b, c, d), then 25% is (a), 50% is (a, b), 75% is (a, b, c), and 100% is (a, b, c, d).

5 Conclusion

In this paper, we have given a generalized quantum circuit using n qubits for the quantum system environment interaction model presented in the quantum Darwinism principle and, using this quantum circuit, the quantum Darwinism state for $n = 2, 3, 4, 5$, and 6 is calculated. The quantum Darwinism principle explains the objectivity of the world, which is fundamentally made out of particles that obey quantum properties. Our experiment test this quantum Darwinism principle on the real quantum devices named **ibmq_athens** and **ibmq_16_melbourne** available on IBM cloud. For this, we have performed the quantum state tomography technique that reconstructs the density matrix from the probabilistic outcomes obtained from the execution of quantum circuits in different Pauli basis settings on the IBM cloud-based superconducting computer. Moreover, the fidelity and purity are calculated to check the quality of reconstructed density matrices, and the quantum mitigation procedure is performed to enhance the quality. Finally, the mutual information, Holevo quantity, and quantum discord are calculated to analyze the information between quantum system and environment.

From the calculated different information, we conclude that to measure the classical information about the state of the quantum system, one did not need to observe the whole interacted environment; just a fragment of it is sufficient. However, he must have all of the information presents in the environment to obtain quantum information delivered by the quantum system. Both these conclusions are according to the theoretical ones. Although From the experimental results, this explanation holds for n = 2 and 3. because, for the rest, the introduced noise and gate error increase while execution and further increase as the n-value increases.

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Author Contributions RS and BKB conceived this research and designed the circuit; RS carried out the experiments and interpret the data; RS wrote the code for experiment and drew all figures and tables. All authors read and approved the final manuscript.

Data Availability The code base created to run these simulations and the related supplementary data could be made available to any reader upon reasonable request.

Declarations

Conflict of interest The authors have no conflict financial and non-financial interest.

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